Mathematical and Computational Methods

Methods

Part I: Foundations and complex numbers (Units 1–3)

Most problems on the exercise sheets are instances of a few standard types, and each type has a *method*: a fixed sequence of steps, taken from the notes, that solves every problem of that type. This guide collects them, unit by unit. Each method states the problem it solves, lists its steps and works one example that is not on any sheet. It ends with the mistakes to avoid, where the notes teach it, and a link to a Review set of problems of that kind. The units fall into four parts: foundations and complex numbers, vectors and matrices, differentiation, and integration.

The guide does not say which sheet problem needs which method. Recognising the type is part of what the sheets train, and in an exam nothing tells you. Where a unit's problems look alike but need different methods, its chapter opens with a table for choosing.

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Part I Foundations and complex numbers

Units 1–3. The first unit's methods are working tools for the whole course: solving an inequality, finding a domain, inverting a function, solving an equation. Units 2 and 3 carry them into the complex numbers, where most problems begin by choosing a form: Cartesian for sums, polar or exponential for products, powers and roots.

Unit 1 Foundations: numbers, sets and functions

Six methods cover the unit's computations. Its other problems (sums in Σ notation, hyperbolic identities, reading a piecewise definition, the irrationality proof) are single moves, done directly as the notes do them.

1.1 Solving an inequality by sign analysis

Problem. Solve an inequality in one real variable, and give the solution set as an interval or a union of intervals.

Method

1. Bring everything to one side, $F(x) > 0$ (or $<$, $\leq$, $\geq$). For a quotient, combine into a single fraction.
2. Factor the numerator and the denominator into linear factors. A factor with no real zero, such as $x^2 + 1$, has one sign everywhere: note its sign.
3. Mark every zero of the numerator and of the denominator on a number line. Between consecutive marked points F keeps one sign: find it from a test point, or by counting negative factors. For $\frac{x-a}{x-b}$ this is the notes' rule for $(x-a)(x-b)$: negative exactly between a and b, positive outside.
4. Keep the intervals with the required sign. Include a zero of the numerator only for $\leq$ or $\geq$. Never include a zero of the denominator.
5. Write the answer in interval notation: a square bracket includes the endpoint, a round one excludes it, and $\pm\infty$ always takes a round one.

Example. Solve $\dfrac{6}{x} \leq x - 1$.

Subtract $\frac{6}{x}$ from both sides and combine:

$$x - 1 - \frac{6}{x} = \frac{x^2 - x - 6}{x} = \frac{(x-3)(x+2)}{x} \geq 0.$$

Call the left-hand side $F(x)$. The numerator is zero at $x = -2$ and $x = 3$, the denominator at

$x = 0$, and the signs of the three factors fix the sign of F on each interval between them:

	$x+2$	x	$x-3$	$F(x)$
$x < -2$	$-$	$-$	$-$	$-$
$-2 < x < 0$	$+$	$-$	$-$	$+$
$0 < x < 3$	$+$	$+$	$-$	$-$
$x > 3$	$+$	$+$	$+$	$+$

We need ≥ 0: keep the two positive intervals, include $x = -2$ and $x = 3$ (zeros of the numerator) and exclude $x = 0$:

$$\boxed{[-2, 0) \cup [3, \infty)}$$

Check. One test point in each interval, in the original inequality: $x = -1$ gives $-6 \leq -2$ and $x = 4$ gives $\frac{3}{2} \leq 3$, both true; $x = -3$ gives $-2 \leq -4$ and $x = 1$ gives $6 \leq 0$, both false. Then the endpoints: at $x = -2$ and $x = 3$ the two sides are equal, so both are included; at $x = 0$ the left side is undefined, so it is excluded.

Variants. A *linear* inequality is solved directly, reversing the sign whenever you multiply or divide by a negative number. A *modulus* is removed first, for $a > 0$: $|x| \leq a \iff -a \leq x \leq a$ and $|x| \geq a \iff x \leq -a$ or $x \geq a$.

Pitfalls

- Multiplying both sides by x, whose sign you do not know. Above, it gives $6 \leq x^2 - x$, so $x \leq -2$ or $x \geq 3$: the interval $(-2, 0)$ is lost and $(-\infty, -2)$ wrongly gained.
- Including a zero of the denominator because the inequality is $\geq$.
- Writing $[3, \infty]$: ∞ is not a number, so it cannot be included.

In the notes. Unit 1: Inequalities, and Unit 1: Intervals and infinity.

Practise it. the Review set on this method

1.2 Maximal domain, and range by building up

Problem. A function is given only by a formula. Find its maximal domain, the largest set of real inputs for which the formula makes sense. Find its range, the set of values it actually takes.

Method

1. List every restriction the formula imposes: a denominator $\neq 0$; the expression under an even root ≥ 0; the expression inside a logarithm > 0; the input of arcsin or arccos in $[-1, 1]$.
2. Solve each restriction (an inequality: Method 1.1).
3. Intersect the solution sets, and write the domain in set-builder notation and as an interval or a union.
4. For the range, first rewrite the formula so that x appears only once, as in $\frac{x+1}{x} = 1 + \frac{1}{x}$. Start from the innermost piece and the set of values it takes as x runs over the domain.
5. Push that set through each operation in turn: a shift, a stretch, a reciprocal, a root. Each must be increasing or decreasing on the current set, so that endpoints go to endpoints. A decreasing step (a minus sign, a reciprocal of positive numbers) reverses the order of the

endpoints.
6. Say which endpoint values are attained and which are only approached.

Example. Find the maximal domain of $f(x) = \dfrac{\ln(5-x)}{\sqrt{x+2}}$, and the range of $g(x) = 1 + \dfrac{4}{x^2+2}$.

For f, the logarithm needs $5 - x > 0$, so $x < 5$. The root needs $x + 2 \geq 0$, and since it is in the denominator it also must not vanish: $x + 2 > 0$, so $x > -2$. Intersecting,

$$\boxed{\{\, x \in \mathbb{R} \;:\; -2 < x < 5 \,\} = (-2, 5)}$$

For g the domain is $\mathbb{R}$. Build up from the inside:

$$x^2 \in [0, \infty) \implies x^2 + 2 \in [2, \infty) \implies \frac{4}{x^2+2} \in (0, 2] \implies g(x) \in (1, 3].$$

The reciprocal reverses the order: the smallest denominator, 2, gives the largest value, 2. The value 3 is attained at $x = 0$; the value 1 is approached as $|x| \to \infty$ but never reached. The range is $\boxed{(1, 3]}$.

Check. Test a value just outside each claimed endpoint: $f(-2)$ divides by zero and $f(5)$ takes $\ln 0$; $g(x) = 1$ would need $\frac{4}{x^2+2} = 0$, which has no solution, and $g(x) > 3$ would need $x^2 + 2 < 2$, which has none either.

Pitfalls

- An even root in a denominator needs the expression under it strictly positive, not just non-negative.
- Simplifying before reading off the domain: $\dfrac{x^2-1}{x-1}$ simplifies to $x + 1$, but $x = 1$ is still excluded.
- Giving the codomain as the range. The range is what is attained.

In the notes. Unit 1: What a function is, with its two worked examples, and Unit 1: Power, polynomial and rational functions.

Practise it. the Review set on this method

1.3 Composition, decomposition and the inverse function

Problem. Form a composite $f \circ g$ with its domain; write a given function as a composite of simpler ones; find the inverse of a function, with its domain.

Method

1. Composite: $(f \circ g)(x) = f\big(g(x)\big)$. Substitute $g(x)$ for the variable of f.
2. Its domain: the x in the domain of g for which $g(x)$ lies in the domain of f. Decide it *before* simplifying the formula.
3. Decomposition: name the outer function $F(u)$, the last operation applied, with a fresh letter u, and the inner function $G(x)$, the whole expression it is applied to.
4. Inverse: show f is injective, usually by showing it is strictly monotonic on its domain, and find its range.

5. Write $y = f(x)$ and solve for x in terms of y.
6. Rename the variable to write $f^{-1}(x)$. The domain of f^{-1} is the range of f, and its range is the domain of f.

Example. Let $f(x) = \sqrt{x}$ and $g(x) = 4 - x$. Find $f \circ g$ and $g \circ f$ with their domains, and the inverse of $h = g \circ f$. Then decompose $p(x) = \mathrm{e}^{-x^2/2}$.

$(f \circ g)(x) = \sqrt{4 - x}$, defined when $4 - x \geq 0$: domain $(-\infty, 4]$. And $(g \circ f)(x) = 4 - \sqrt{x}$, defined when x is in the domain of f: domain $[0, \infty)$.

$h(x) = 4 - \sqrt{x}$ is strictly decreasing on $[0, \infty)$, since $\sqrt{x}$ is strictly increasing. Its range is $(-\infty, 4]$, with 4 attained at $x = 0$. Solve $y = 4 - \sqrt{x}$: $\sqrt{x} = 4 - y$, so $x = (4 - y)^2$. Hence

$$\boxed{h^{-1}(x) = (4 - x)^2}$$

with domain $(-\infty, 4]$, the range of h, and range $[0, \infty)$, the domain of h. For p, the last operation is raising e to a power, applied to $-x^2/2$: $G(x) = -\frac{1}{2}x^2$ and $F(u) = \mathrm{e}^u$, so $p = F \circ G$.

Check. $h\big(h^{-1}(1)\big) = 4 - \sqrt{9} = 1$ and $h^{-1}\big(h(9)\big) = h^{-1}(1) = 9$.

Pitfalls

- Reading the domain off the simplified formula. With $g(x) = \sqrt{x}$ and $f(x) = x^2$, $(f \circ g)(x) = x$, but only for $x \geq 0$.
- Dropping the domain of the inverse. $(4 - x)^2$ on all of $\mathbb{R}$ is not injective; it inverts h only on $(-\infty, 4]$.
- Writing f^{-1} for $1/f$. They are different functions.

In the notes. Unit 1: Composition of functions, and Unit 1: Inverse functions and monotonicity.
Practise it. the Review set on this method

1.4 Testing a property: injective, surjective, even, odd, periodic

Problem. Decide whether $f\colon A \to B$ is injective, surjective or bijective; whether it is even, odd or neither; and find its period.

Method

1. Injective: show that f is strictly monotonic on A, or that $f(x_1) = f(x_2)$ forces $x_1 = x_2$. To refute it, give $x_1 \neq x_2$ with $f(x_1) = f(x_2)$.
2. Surjective: solve $y = f(x)$ for $x \in A$, for every $y \in B$; equivalently, show that the range is all of B. To refute it, give one $y \in B$ that is never attained.
3. Even or odd: check that the domain is symmetric about 0. Compute $f(-x)$, simplify, and compare it with $f(x)$ and with $-f(x)$.
4. Neither: give one x with $f(-x) \neq f(x)$ and one with $f(-x) \neq -f(x)$. A single x can do both.
5. Period: for a non-constant f, find the smallest $T > 0$ with $f(x + T) = f(x)$ for all x, if there is one. If f has period T, then $f(bx)$ has period T/b for $b > 0$.

Example. Let $f\colon \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{x}{1+x^2}$. Is it even or odd, injective, surjective? And what is the period of $q(x) = \cos\dfrac{\pi x}{2}$?

$f(-x) = \dfrac{-x}{1+x^2} = -f(x)$ on the symmetric domain $\mathbb{R}$, so f is *odd*.

Solve $y = f(x)$ for x: $yx^2 - x + y = 0$. For $y \neq 0$ this quadratic has discriminant $1 - 4y^2$, so it has a real root only when $|y| \leq \frac{1}{2}$ (and $y = 0$ gives $x = 0$). The range is $[-\frac{1}{2}, \frac{1}{2}]$, so $y = 1$ is never attained: f is *not surjective*.

For $0 < |y| < \frac{1}{2}$ there are two roots. Taking $y = \frac{2}{5}$ gives $2x^2 - 5x + 2 = 0$, so $x = 2$ or $x = \frac{1}{2}$: $f(2) = \frac{2}{5} = f(\frac{1}{2})$, and f is *not injective*.

cos has period 2π, so q has period $2\pi \,/\, \frac{\pi}{2} = \boxed{4}$.

Check. $q(x+4) = \cos\big(\frac{\pi x}{2} + 2\pi\big) = q(x)$, and no smaller T works: $q(T) = q(0) = 1$ needs $\frac{\pi T}{2}$ to be a multiple of 2π, so T is a multiple of 4. And $f(\pm 1) = \pm\frac{1}{2}$, the ends of the range, are attained.

Pitfalls

- Judging on the wrong domain or codomain. The verdict belongs to the rule *together with* A and B: x^2 is bijective as a map $[0,\infty) \to [0,\infty)$.
- Proving evenness from one value. $f(-x) = f(x)$ must hold for every x; a single value can only refute it.
- Calling a function even or odd when its domain is not symmetric: $\sqrt{x}$ is neither.

In the notes. Unit 1: Injective, surjective and bijective functions, and Unit 1: Symmetry: even, odd and periodic functions.

Practise it. the Review set on this method

1.5 Transforming a graph from a base graph

Problem. Sketch $y = a\,f\big(b(x-c)\big) + d$ from a graph you know, or name the transformations that produce a given formula.

Method

1. Write the target in the form $a\,f\big(b(x-c)\big)+d$: complete the square, or factor the coefficient of x out of the bracket.
2. Every point (u, v) of $y = f(x)$ moves to $\big(c + \frac{u}{b},\ d + av\big)$. Horizontally: scale by $\frac{1}{b}$ (reflecting if $b < 0$), then shift by c. Vertically: scale by a (reflecting if $a < 0$), then shift by d.
3. Move the key points and the asymptotes by that rule. A moved intercept need not be an intercept of the new graph: find those from its formula, with $x = 0$ and $y = 0$.
4. Read off what is asked: the domain, the range, where the function increases.
5. Sketch, labelling the points you moved.

Example. Obtain $y = 1 + \sqrt{4 - 2x}$ from $y = \sqrt{x}$, and give its domain and range.

Factor the coefficient of x out of the bracket: $4 - 2x = -2(x-2)$, so the target is $f\big(b(x-c)\big)+d$ with $f = \sqrt{}$, $a = 1$, $b = -2$, $c = 2$, $d = 1$. The graph is compressed horizontally by $\frac{1}{2}$ and

reflected in the y-axis, then shifted 2 right and 1 up. The point (u, v) goes to $\left(2 - \frac{u}{2},\ 1 + v\right)$:

$$(0,0) \mapsto (2,1), \qquad (4,2) \mapsto (0,3), \qquad (16,4) \mapsto (-6,5).$$

The graph starts at $(2,1)$ and rises to the *left*, so

$$\boxed{\text{domain } (-\infty, 2], \quad \text{range } [1, \infty)}$$

and the function is decreasing. It meets the y-axis at $(0,3)$ and never meets the x-axis, since $y \geq 1$.

Check. Substitute the moved points: $1 + \sqrt{4-0} = 3$ and $1 + \sqrt{4+12} = 5$. The domain also follows from $4 - 2x \geq 0$.

Pitfalls

- Reading $f(2x-4)$ as a shift by 4. Factor first: $f\big(2(x-2)\big)$ is a shift by 2.
- Moving the wrong way: $f(x-c)$ moves the graph *right* by c when $c > 0$.
- Doing the vertical steps in the wrong order: $af(x)+d$ scales first, then shifts; $a\big(f(x)+d\big)$ is a different graph.

In the notes. Unit 1: Graphs and graph transformations, and its table of transformations.
Practise it. the Review set on this method

1.6 Exponential, logarithmic and trigonometric equations

Problem. Solve an equation in which the unknown sits inside an exponential, a logarithm or a trigonometric function.

Method

1. If a logarithm of an expression in x appears, find where it is defined first.
2. Use the index and log laws to reduce the equation to one exponential or one logarithm on each side, or substitute $u = \mathrm{e}^x$ (so $u > 0$) to get a polynomial in u.
3. Take ln of both sides, or make each side a power of e, and solve. After a substitution, solve the polynomial for u, keep only the roots with $u > 0$, and set $x = \ln u$.
4. Reject any value outside the domain from step 1.
5. For a trigonometric equation, isolate $\sin\theta = c$, $\cos\theta = c$ or $\tan\theta = c$, where θ is the whole argument (such as $2x$), factoring first if there are several terms.
6. Take the principal value α from the table or the inverse function. The second value in one period comes from the symmetry: $\pi - \alpha$ for sine, $-\alpha$ for cosine; tangent has only one value per period.
7. Add whole periods to θ (2π for sine and cosine, π for tangent), then solve for x, and list every solution in the interval asked for.

Example. Solve (a) $\ln(x+4) = 2\ln x$, and (b) $2\sin^2 x = \sin x$ on $[0, 2\pi)$.

(a) Both logarithms need $x > 0$ (then $x + 4 > 0$ too). By the power law $2\ln x = \ln(x^2)$, so $\ln(x+4) = \ln(x^2)$, $x + 4 = x^2$ and $x^2 - x - 4 = 0$, with roots $x = \frac{1 \pm \sqrt{17}}{2}$. Only the plus sign gives $x > 0$: $\boxed{x = \frac{1+\sqrt{17}}{2}}$.

(b) Bring everything to one side and factor: $\sin x\,(2\sin x - 1) = 0$. Then $\sin x = 0$ gives $x = 0,\ \pi$. And $\sin x = \frac{1}{2}$ gives the principal value $\frac{\pi}{6}$ and the second value $\pi - \frac{\pi}{6} = \frac{5\pi}{6}$. So

$$\boxed{x \in \left\{0,\ \tfrac{\pi}{6},\ \tfrac{5\pi}{6},\ \pi\right\}}$$

Check. Substitute: in (a), $x^2 = \frac{9+\sqrt{17}}{2} = x + 4$ and $x > 0$; in (b), $2\sin^2 x = \sin x$ holds at each of the four values ($\frac{1}{2} = \frac{1}{2}$ at $\frac{\pi}{6}$ and $\frac{5\pi}{6}$). Then count: each factor, $\sin x = 0$ and $\sin x = \frac{1}{2}$, has two solutions in $[0, 2\pi)$, so there are four.

Pitfalls

- Dividing by $\sin x$: it loses the solutions $x = 0$ and $x = \pi$. Factor instead, or check the zeros of what you divide by.
- Keeping $x = \frac{1-\sqrt{17}}{2}$ in (a). It satisfies $\ln(x + 4) = \ln(x^2)$, but $\ln x$ is not defined there: $2\ln x = \ln(x^2)$ holds only for $x > 0$.
- Stopping at the principal value. $\arcsin\frac{1}{2}$ is one number; the equation $\sin x = \frac{1}{2}$ has two solutions in every period.

In the notes. Unit 1: Exponential and logarithmic functions, and Unit 1: Trigonometric and inverse-trigonometric functions.

Practise it. the Review set on this method

Unit 2 Complex numbers I: algebra and geometry

Six methods. The powers of i are a step inside the first, and the short proofs of the conjugate and modulus rules are the last step of Method 2.5.

2.1 Cartesian arithmetic, and division by the conjugate

Problem. Write a sum, product, power or quotient of complex numbers in the form $a + b\,\mathrm{i}$; find a conjugate or a modulus.

Method

1. Sum or difference: real parts with real parts, imaginary parts with imaginary parts.
2. Product: expand the brackets, replace i^2 by -1, and group into $(\dots) + (\dots)\,\mathrm{i}$. For i^n, divide n by 4: $\mathrm{i}^n = \mathrm{i}^r$, where r is the remainder, read from $1,\ \mathrm{i},\ -1,\ -\mathrm{i}$. For $n < 0$, first add a multiple of 4: $\mathrm{i}^{-5} = \mathrm{i}^{-5+8} = \mathrm{i}^3 = -\mathrm{i}$.
3. Quotient: multiply the top and the bottom by the conjugate $c - d\,\mathrm{i}$ of the denominator $c + d\,\mathrm{i}$. The denominator becomes the real number $c^2 + d^2$.
4. Expand the numerator, replace i^2 by -1, and split into $\dfrac{p}{c^2+d^2} + \dfrac{q}{c^2+d^2}\,\mathrm{i}$.
5. Conjugate: flip the sign of the imaginary part. Modulus: $|a + b\,\mathrm{i}| = \sqrt{a^2 + b^2}$, and $z\bar{z} = |z|^2$.

Example. Write $\dfrac{7+\mathrm{i}}{1-2\mathrm{i}} - \mathrm{i}^{23}$ in the form $a + b\,\mathrm{i}$, and find its modulus.

Multiply the top and the bottom by $1 + 2\mathrm{i}$, the conjugate of the denominator:

$$\frac{7+\mathrm{i}}{1-2\mathrm{i}} = \frac{(7+\mathrm{i})(1+2\mathrm{i})}{(1-2\mathrm{i})(1+2\mathrm{i})} = \frac{7+14\mathrm{i}+\mathrm{i}+2\mathrm{i}^2}{1^2+2^2} = \frac{5+15\mathrm{i}}{5} = 1+3\mathrm{i}.$$

Since $23 = 4 \cdot 5 + 3$, $\mathrm{i}^{23} = \mathrm{i}^3 = -\mathrm{i}$. So the number is $1 + 3\mathrm{i} - (-\mathrm{i}) = \boxed{1+4\mathrm{i}}$, with modulus $\sqrt{1^2+4^2} = \boxed{\sqrt{17}}$.

Check. Multiply the quotient back by the denominator: $(1+3\mathrm{i})(1-2\mathrm{i}) = 1 - 2\mathrm{i} + 3\mathrm{i} - 6\mathrm{i}^2 = 7 + \mathrm{i}$. Then $\mathrm{i}^{23} = -\mathrm{i}$ because $(-\mathrm{i})\,\mathrm{i} = 1 = \mathrm{i}^{24}$, and $(1+4\mathrm{i})(1-4\mathrm{i}) = 1 - 16\mathrm{i}^2 = 17 = |1+4\mathrm{i}|^2$.

Pitfalls

- Dividing part by part: $\dfrac{a+b\mathrm{i}}{c+d\mathrm{i}} \neq \dfrac{a}{c} + \dfrac{b}{d}\,\mathrm{i}$.
- Losing the sign of i^2: $2\mathrm{i} \cdot \mathrm{i} = -2$, not 2.
- Putting i inside the modulus: $|1 + 4\mathrm{i}| = \sqrt{1+16}$, not $\sqrt{1+(4\mathrm{i})^2}$.

In the notes. Unit 2: The imaginary unit, for the powers of i, Unit 2: Addition, subtraction and multiplication, Unit 2: The modulus, and Unit 2: Division.

Practise it. the Review set on this method

2.2 Real polynomials: complex roots in conjugate pairs

Problem. Solve a real quadratic with a negative discriminant; given one non-real root of a real cubic or quartic, find all its roots.

Method

1. Quadratic: complete the square to $(x+p)^2+q=0$ with $q>0$. Then $(x+p)^2=-q=(\mathrm{i}\sqrt{q})^2$, so $x=-p\pm\mathrm{i}\sqrt{q}$. Or use the formula: $D=b^2-4ac<0$ and $x=\dfrac{-b\pm\mathrm{i}\sqrt{4ac-b^2}}{2a}$.
2. A polynomial with *real* coefficients and a root α also has the root $\overline{\alpha}$.
3. Multiply the pair into a real quadratic: $(x-\alpha)(x-\overline{\alpha})=x^2-2\,\mathrm{Re}(\alpha)\,x+|\alpha|^2$.
4. Find the other factor by comparing coefficients (or by long division). Its roots (by step 1, if it is a quadratic) are the remaining roots.
5. List every root.

Example. (a) Solve $x^2-6x+13=0$. (b) Given that $-1+2\mathrm{i}$ is a root of $p(x)=2x^3+x^2+4x-15$, find the other two.

(a) Complete the square: $(x-3)^2+4=0$, so $(x-3)^2=-4=(2\mathrm{i})^2$ and $\boxed{x=3\pm2\mathrm{i}}$.

(b) The coefficients are real, so $-1-2\mathrm{i}$ is a root too. The pair gives the real factor

$$x^2-2\cdot(-1)\,x+|-1+2\mathrm{i}|^2=x^2+2x+5.$$

The x^3 terms need $2x$ in the other factor, so write $p(x)=(x^2+2x+5)(2x+k)$ and compare the constant terms: $5k=-15$, so $k=-3$. The roots are $\boxed{-1+2\mathrm{i},\ -1-2\mathrm{i},\ \tfrac{3}{2}}$.

Check. Expand: $(x^2+2x+5)(2x-3)=2x^3+x^2+4x-15$. In (a), the pair multiplies back to $x^2-2\cdot3\,x+|3+2\mathrm{i}|^2=x^2-6x+13$.

Pitfalls

- Dropping the i: from $(x-3)^2=-4$ it is $x-3=\pm2\mathrm{i}$, because $(2\mathrm{i})^2=-4$. Do not write $\sqrt{-4}$.
- Using the conjugate-pair rule on a polynomial with non-real coefficients: $x-\mathrm{i}$ has the root i but not $-\mathrm{i}$.
- Stopping at the pair: a cubic has a third root, and it is real.

In the notes. Unit 2: Quadratic equations with complex roots.

Practise it. the Review set on this method

2.3 Cartesian and polar form, with the quadrant rule

Problem. Write a nonzero $z=a+b\,\mathrm{i}$ in polar form $r(\cos\theta+\mathrm{i}\sin\theta)$ with its principal argument, or convert a polar form back to $a+b\,\mathrm{i}$.

Method

1. $r=|z|=\sqrt{a^2+b^2}$.
2. Sketch the point to see its quadrant. A point on an axis has $\theta=0$, $\frac{\pi}{2}$, π or $-\frac{\pi}{2}$ directly.
3. The reference angle is the acute angle $\alpha=\arctan|b/a|$.

4. The principal argument, in $(-\pi, \pi]$, from the quadrant: α (first), $\pi - \alpha$ (second), $-\pi + \alpha$ (third), $-\alpha$ (fourth).
5. Write $z = r(\cos\theta + \mathrm{i}\sin\theta)$.
6. Back to Cartesian: evaluate $\cos\theta$ and $\sin\theta$, and multiply by r.

Example. Write $z = -3 - \sqrt{3}\,\mathrm{i}$ in polar form with its principal argument, and $w = 5\big(\cos\frac{3\pi}{4} + \mathrm{i}\sin\frac{3\pi}{4}\big)$ in Cartesian form.

$r = \sqrt{9+3} = 2\sqrt{3}$. Both parts are negative: the third quadrant. The reference angle is $\alpha = \arctan\frac{\sqrt{3}}{3} = \frac{\pi}{6}$, so $\theta = -\pi + \frac{\pi}{6} = -\frac{5\pi}{6}$ and

$$\boxed{z = 2\sqrt{3}\Big(\cos\big(-\tfrac{5\pi}{6}\big) + \mathrm{i}\sin\big(-\tfrac{5\pi}{6}\big)\Big)}$$

For w: $\cos\frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$ and $\sin\frac{3\pi}{4} = \frac{\sqrt{2}}{2}$, so $\boxed{w = -\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}\,\mathrm{i}}$.

Check. Convert back: $2\sqrt{3}\big(-\frac{\sqrt{3}}{2} - \frac{1}{2}\mathrm{i}\big) = -3 - \sqrt{3}\,\mathrm{i}$.

Pitfalls

- Taking $\arctan(b/a)$ as the argument. Here it gives $\arctan\frac{\sqrt{3}}{3} = \frac{\pi}{6}$, a first-quadrant angle for a third-quadrant point. The sketch fixes the quadrant, not the tangent.
- Giving $\frac{7\pi}{6}$ when the principal argument is asked for. It is an argument of z, but not in $(-\pi, \pi]$.
- Dividing by $a = 0$. A point on the imaginary axis needs no tangent.

In the notes. Unit 2: Modulus and argument, and Unit 2: Converting between Cartesian and polar form.

Practise it. the Review set on this method

2.4 Multiplying and dividing in polar form; rotation

Problem. Multiply or divide complex numbers given in polar form; rotate and scale a point of the plane.

Method

1. Write each factor in polar form (Method 2.3).
2. Product: multiply the moduli and add the arguments. Quotient: divide the moduli and subtract the arguments.
3. If the principal argument is asked for and the result is outside $(-\pi, \pi]$, add or subtract 2π.
4. Convert to $a + b\,\mathrm{i}$ if asked.
5. To turn a point z through α anticlockwise and scale it by $k > 0$, multiply by $k(\cos\alpha + \mathrm{i}\sin\alpha)$. A clockwise turn uses $-\alpha$.

Example. Let $z = 2\big(\cos(-\frac{2\pi}{3}) + \mathrm{i}\sin(-\frac{2\pi}{3})\big)$ and $w = 4\big(\cos\frac{\pi}{2} + \mathrm{i}\sin\frac{\pi}{2}\big)$. Find zw and z/w in Cartesian form. Then turn $2 + 4\mathrm{i}$ a quarter turn clockwise and halve its distance from 0.

zw has modulus 8 and argument $-\frac{2\pi}{3} + \frac{\pi}{2} = -\frac{\pi}{6}$, already in $(-\pi, \pi]$. So

$$zw = 8\Big(\frac{\sqrt{3}}{2} - \frac{1}{2}\,\mathrm{i}\Big) = \boxed{4\sqrt{3} - 4\mathrm{i}}.$$

z/w has modulus $\frac{1}{2}$ and argument $-\frac{2\pi}{3} - \frac{\pi}{2} = -\frac{7\pi}{6}$, which is outside $(-\pi, \pi]$; add 2π to get $\frac{5\pi}{6}$: $z/w = \frac{1}{2}\big(-\frac{\sqrt{3}}{2} + \frac{1}{2}\mathrm{i}\big) = \boxed{-\frac{\sqrt{3}}{4} + \frac{1}{4}\,\mathrm{i}}$.

The multiplier has $k = \frac{1}{2}$ and $\alpha = -\frac{\pi}{2}$, so it is $\frac{1}{2}(\cos(-\frac{\pi}{2}) + \mathrm{i}\sin(-\frac{\pi}{2})) = -\frac{1}{2}\mathrm{i}$. Then $(2 + 4\mathrm{i})(-\frac{1}{2}\mathrm{i}) = -\mathrm{i} - 2\mathrm{i}^2 = \boxed{2 - \mathrm{i}}$.

Check. Undo the turn and the scaling, multiplying by 2i: $(2 - \mathrm{i}) \cdot 2\mathrm{i} = 4\mathrm{i} - 2\mathrm{i}^2 = 2 + 4\mathrm{i}$. And zw is $2 \cdot 4 = 8$ from 0: $\sqrt{48 + 16} = 8$.

Pitfalls

- Adding the moduli. They multiply; only the arguments add.
- Leaving $-\frac{7\pi}{6}$ as the principal argument.
- Turning the wrong way: $\cos\alpha + \mathrm{i}\sin\alpha$ with $\alpha > 0$ turns anticlockwise.

In the notes. Unit 2: Multiplication and division in polar form, and Unit 2: A general product as rotation and scaling.

Practise it. the Review set on this method

2.5 Equating real and imaginary parts

Problem. Find an unknown complex number from an equation it satisfies; prove an identity that holds for all complex numbers.

Method

1. Write the unknown as $z = x + y\,\mathrm{i}$ with x and y real (and $\overline{z} = x - y\,\mathrm{i}$).
2. Expand both sides into (real) + (real) i.
3. Two complex numbers are equal exactly when their real parts are equal and their imaginary parts are equal. That gives two real equations.
4. Solve the two equations for x and y.
5. For an identity in z and w, write $z = a + b\,\mathrm{i}$ and $w = c + d\,\mathrm{i}$, expand both sides and compare the parts. Where moduli appear, it is often shorter to use $|u|^2 = u\,\overline{u}$ with the conjugate rules $\overline{z \pm w} = \overline{z} \pm \overline{w}$ and $\overline{zw} = \overline{z}\,\overline{w}$.

Example. Find every z with $2z + \mathrm{i}\overline{z} = 3 + 6\mathrm{i}$.

Put $z = x + y\,\mathrm{i}$, so $\overline{z} = x - y\,\mathrm{i}$, and expand:

$$2x + 2y\,\mathrm{i} + \mathrm{i}(x - y\,\mathrm{i}) = (2x + y) + (x + 2y)\,\mathrm{i}.$$

Equate the parts with those of $3 + 6\mathrm{i}$: $2x + y = 3$ and $x + 2y = 6$. The first gives $y = 3 - 2x$;

then $x + 6 - 4x = 6$, so $x = 0$ and $y = 3$. The only solution is $\boxed{z = 3\mathrm{i}}$.

Check. Substitute: $2(3\mathrm{i}) + \mathrm{i}(-3\mathrm{i}) = 6\mathrm{i} + 3$.

Pitfalls

- Treating $\overline{z}$ as z: they have the same real part and opposite imaginary parts.
- Forgetting that x and y must be real. If the two equations have no real solution, no z satisfies the equation.
- Checking an identity at one value of z. That can refute it but not prove it.

In the notes. Unit 2: The imaginary unit, for when two complex numbers are equal, and Unit 2: Why the modulus behaves well, for the $|u|^2 = u\overline{u}$ proofs.

Practise it. the Review set on this method

2.6 Reading a locus on the Argand plane

Problem. Describe and sketch a set of points z given by distances: a circle $|z - c| = r$, a disc such as $|z - c| < r$, or a line $|z - a| = |z - b|$.

Method

1. Read $|z - c|$ as the distance from z to the point c. Rewrite $|z + c|$ as $|z - (-c)|$ first.
2. Name the set. $|z - c| = r$ is the circle with centre c and radius r; $|z - c| < r$ is the inside of that circle. $|z - a| = |z - b|$ is the perpendicular bisector of the segment from a to b.
3. If you are unsure, put $z = x + y\,\mathrm{i}$, square both sides, and simplify to an equation (or inequality) in x and y.
4. Sketch it. For a circle, mark the centre and the radius; for a bisector, the two points. A boundary that belongs to the set is drawn solid; one that does not (a strict inequality) is drawn dashed.

Example. Describe (a) $|z - 1 + 2\mathrm{i}| = 3$ and (b) $|z + 2\mathrm{i}| = |z - 4|$.

(a) $|z - (1 - 2\mathrm{i})| = 3$: the points at distance 3 from $1 - 2\mathrm{i}$, that is, the circle

$$\boxed{\text{centre } (1, -2),\ \text{radius } 3}$$

(b) $|z - (-2\mathrm{i})| = |z - 4|$: the points as far from $-2\mathrm{i}$ as from 4, that is, the perpendicular bisector of the segment from $(0, -2)$ to $(4, 0)$. With $z = x + y\,\mathrm{i}$ and both sides squared,

$$x^2 + (y + 2)^2 = (x - 4)^2 + y^2 \iff 4y + 4 = -8x + 16 \iff \boxed{y = 3 - 2x}.$$

Check. The midpoint $(2, -1)$ of the segment lies on $y = 3 - 2x$, and the slope -2 is perpendicular to the segment's slope $\frac{1}{2}$.

Pitfalls

- Reading $|z + 2\mathrm{i}|$ as the distance from $2\mathrm{i}$. It is the distance from $-2\mathrm{i}$.
- Getting the centre's sign wrong: $|z - 1 + 2\mathrm{i}|$ is centred at $1 - 2\mathrm{i}$, not at $1 + 2\mathrm{i}$.
- Drawing the boundary of $|z - c| < r$ solid.

In the notes. Unit 2: Addition and subtraction as vectors, where $|z - w|$ is the distance between z and w.

Practise it. the Review set on this method

Unit 3 Complex numbers II: exponential form, De Moivre, roots

Five methods, and the problems look alike: most start by writing a number as $re^{i\theta}$. What decides the method is what the question asks for. The fundamental theorem of algebra is a fact to quote; given one non-real root, a real polynomial is factored as in Method 2.2.

Which method?

The question asks for	Method
a product, a quotient or a rotation	3.1, exponential form and index laws
an integer power z^n	3.2, De Moivre (not the binomial theorem)
every w with $w^n = z$; the roots of unity	3.3, nth roots
a square root of z when $\arg z$ is not a standard angle	3.3, variant: equate parts
$\cos n\theta$ or $\sin n\theta$ in powers of $\cos\theta$ and $\sin\theta$	3.4, expand $(\cos\theta + i\sin\theta)^n$
$\cos^n\theta$ or $\sin^n\theta$ as a sum of multiple angles	3.4, the inverse Euler formulas
cos, sin, cosh or sinh of an imaginary number	3.4, the inverse Euler formulas at ix
one sinusoid equal to a sum of sinusoids of one frequency	3.5, phasors

3.1 Exponential form and the index laws

Problem. Write a nonzero complex number as $re^{i\theta}$ or back as $a + bi$; multiply, divide, invert or rotate in exponential form.

Method

1. To exponential form: find r and the argument θ exactly as for the polar form (Method 2.3, quadrant first), and write $z = re^{i\theta}$.
2. Back to Cartesian: $re^{i\theta} = r\cos\theta + ir\sin\theta$.
3. Product and quotient: multiply or divide the moduli, add or subtract the exponents. The reciprocal is $\frac{1}{r}e^{-i\theta}$ and the conjugate is $re^{-i\theta}$.
4. Rotation: multiplying by $e^{i\alpha}$ turns a point through α anticlockwise about 0 ($e^{-i\alpha}$ turns it clockwise), and $se^{i\alpha}$ also scales it by s.
5. If the principal argument is asked for, bring the exponent into $(-\pi, \pi]$ by whole turns.

Example. Let $z = -2 + 2i$ and $w = \sqrt{2} - \sqrt{2}\,i$. Find zw and z/w through the exponential form.

z is in the second quadrant with $|z| = 2\sqrt{2}$ and reference angle $\frac{\pi}{4}$, so $z = 2\sqrt{2}\,e^{i3\pi/4}$. w is in the fourth with $|w| = 2$, so $w = 2e^{-i\pi/4}$. Then

$$zw = 4\sqrt{2}\,e^{i(3\pi/4-\pi/4)} = 4\sqrt{2}\,e^{i\pi/2} = \boxed{4\sqrt{2}\,i}, \qquad \frac{z}{w} = \sqrt{2}\,e^{i(3\pi/4+\pi/4)} = \sqrt{2}\,e^{i\pi} = \boxed{-\sqrt{2}}.$$

Check. In Cartesian form, $(-2 + 2i)(\sqrt{2} - \sqrt{2}i) = -2\sqrt{2} + 2\sqrt{2}i + 2\sqrt{2}i + 2\sqrt{2} = 4\sqrt{2}\,i$, and $(-\sqrt{2})(\sqrt{2} - \sqrt{2}i) = -2 + 2i = z$.

Pitfalls

- Subtracting the exponents the wrong way round in a quotient: it is the argument of the top minus the argument of the bottom.
- Writing $e^{i\theta}$ with θ in degrees. The exponent is in radians.
- Taking the argument from $\arctan(b/a)$ without the quadrant, as in Method 2.3.

In the notes. Unit 3: The exponential form of a complex number, and Unit 3: $e^{i\theta}$ as a rotation.
Practise it. the Review set on this method

3.2 Integer powers by De Moivre

Problem. Compute z^n for an integer n, positive or negative.

Method

1. Write $z = re^{i\theta}$ (Method 3.1).
2. Then $z^n = r^n e^{in\theta}$: raise the modulus to the power and multiply the argument by n. This works for negative n too.
3. Remove whole turns (2π each) from $n\theta$ to bring it into $(-\pi, \pi]$.
4. Evaluate cos and sin of it and write $a + b\,\mathrm{i}$.

Why it works. De Moivre's theorem, $(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$. The proof is by induction: check $n = 1$; assume it for $n = k$; split $(\cos\theta + \mathrm{i}\sin\theta)^{k+1}$ into the kth power times one more factor, use the assumption, and add the arguments.

Example. Compute (a) $(-\sqrt{3} - \mathrm{i})^7$ and (b) $(1 - \sqrt{3}\,\mathrm{i})^{-5}$.

(a) $-\sqrt{3} - \mathrm{i}$ is in the third quadrant with modulus 2 and reference angle $\frac{\pi}{6}$, so $-\sqrt{3} - \mathrm{i} = 2e^{-\mathrm{i}5\pi/6}$ and $(-\sqrt{3} - \mathrm{i})^7 = 2^7 e^{-\mathrm{i}35\pi/6}$. Add three whole turns: $-\frac{35\pi}{6} + 6\pi = \frac{\pi}{6}$. So the power is $128e^{\mathrm{i}\pi/6} = \boxed{64\sqrt{3} + 64\mathrm{i}}$.

(b) $1 - \sqrt{3}\,\mathrm{i} = 2e^{-\mathrm{i}\pi/3}$, so

$$(1 - \sqrt{3}\,\mathrm{i})^{-5} = 2^{-5}e^{\mathrm{i}5\pi/3} = \tfrac{1}{32}e^{-\mathrm{i}\pi/3} = \tfrac{1}{32}\left(\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i}\right) = \boxed{\tfrac{1}{64} - \tfrac{\sqrt{3}}{64}\,\mathrm{i}}.$$

Check. Multiply out a low power. $(-\sqrt{3}-\mathrm{i})^3 = -8\mathrm{i}$, so $(-\sqrt{3}-\mathrm{i})^7 = (-8\mathrm{i})^2(-\sqrt{3}-\mathrm{i}) = 64\sqrt{3}+64\mathrm{i}$. And $(1 - \sqrt{3}\,\mathrm{i})^3 = -8$, so $(1 - \sqrt{3}\,\mathrm{i})^{-5} = (1 - \sqrt{3}\,\mathrm{i})/(1 - \sqrt{3}\,\mathrm{i})^6 = \frac{1}{64}(1 - \sqrt{3}\,\mathrm{i})$.

Pitfalls

- Raising the modulus but not multiplying the argument, or the reverse.
- Expanding by the binomial theorem: it gives the same answer in eight terms instead of one line.
- Losing track of the turns: $-\frac{35\pi}{6}$ is $\frac{\pi}{6}$ less three whole turns, not $-\frac{5\pi}{6}$ less some.

In the notes. Unit 3: Statement and proof, and Unit 3: Integer powers of complex numbers.
Practise it. the Review set on this method

3.3 nth roots, and the roots of unity

Problem. Find every w with $w^n = z$, in exponential or Cartesian form.

Method

1. Write $z = re^{i\theta}$, with the quadrant checked. (If $z = 0$, the only root is $w = 0$.)
2. The roots all have modulus $\rho = r^{1/n}$, the positive real root.
3. Their arguments are $\phi_k = \dfrac{\theta + 2\pi k}{n}$ for $k = 0, 1, \ldots, n-1$: exactly n of them.
4. Write $w_k = \rho\, e^{i\phi_k}$, and convert to Cartesian form if asked. Each root is the previous one turned through $\frac{2\pi}{n}$: $w_k = w_0\, e^{2\pi i k/n}$.
5. The roots are the corners of a regular n-gon on the circle of radius ρ.

Variants. The *roots of unity* are the case $z = 1$: the powers of $\omega = e^{2\pi i/n}$, which add up to 0 for $n \geq 2$. A *square root* of z when $\arg z$ is not a standard angle: put $(a+bi)^2 = z$, equate the parts, and add $a^2 + b^2 = |z|$ from the moduli. Solve for a^2 and b^2, and choose the signs so that $2ab = \operatorname{Im} z$: the roots are $\pm(a+bi)$. An equation such as $z^n = c\,\bar{z}$: treat $z = 0$ separately, then match the moduli and the arguments.

Example. Solve $w^4 = -8 + 8\sqrt{3}\,i$ in Cartesian form.

$|z| = \sqrt{64 + 192} = 16$, and z is in the second quadrant with reference angle $\arctan\sqrt{3} = \frac{\pi}{3}$, so $\theta = \frac{2\pi}{3}$. Then $\rho = 16^{1/4} = 2$ and

$$\phi_k = \frac{2\pi/3 + 2\pi k}{4} = \frac{\pi}{6} + \frac{k\pi}{2}, \qquad k = 0, 1, 2, 3: \quad \tfrac{\pi}{6},\ \tfrac{2\pi}{3},\ \tfrac{7\pi}{6},\ \tfrac{5\pi}{3}.$$

The roots, the corners of a square on the circle of radius 2, are

$$\boxed{\sqrt{3} + i,\quad -1 + \sqrt{3}\,i,\quad -\sqrt{3} - i,\quad 1 - \sqrt{3}\,i}$$

Check. Raise one root to the fourth power: $(\sqrt{3}+i)^4 = (2e^{i\pi/6})^4 = 16e^{i2\pi/3} = -8 + 8\sqrt{3}\,i$. The four roots add up to 0, as the corners of a regular polygon centred at 0 must.

Pitfalls

- Forgetting the $2\pi k$: it gives one root instead of n.
- Running k up to n: $k = n$ repeats the root for $k = 0$.
- Taking $\rho = r$ instead of $r^{1/n}$.

In the notes. Unit 3: The nth roots of a complex number, Unit 3: Square roots the algebraic way, and Unit 3: The roots of unity.

Practise it. the Review set on this method

3.4 Identities from De Moivre and the inverse Euler formulas

Problem. Write $\cos n\theta$ or $\sin n\theta$ in powers of $\cos\theta$ and $\sin\theta$; write a power of $\cos\theta$ or $\sin\theta$ as a sum of multiple angles; evaluate a trigonometric or hyperbolic function at an imaginary number.

Method

1. Multiple angle: expand $(\cos\theta + \mathrm{i}\sin\theta)^n$ by the binomial theorem. By De Moivre it equals $\cos n\theta + \mathrm{i}\sin n\theta$. Compare the real parts and the imaginary parts. Use $\sin^2\theta = 1 - \cos^2\theta$ if one function is asked for.
2. Power: substitute $\cos\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$ or $\sin\theta = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})$, take the factor $(\frac{1}{2})^n$ or $(\frac{1}{2\mathrm{i}})^n$ outside, and expand.
3. Pair each $\mathrm{e}^{\mathrm{i}k\theta}$ with $\mathrm{e}^{-\mathrm{i}k\theta}$: their sum is $2\cos k\theta$, their difference $2\mathrm{i}\sin k\theta$. Simplify.
4. Imaginary argument: put $\mathrm{i}x$ into the inverse Euler formula, use $\mathrm{i}^2 = -1$, and recognise $\cosh x$ or $\sinh x$. So $\cos(\mathrm{i}x) = \cosh x$ and $\sin(\mathrm{i}x) = \mathrm{i}\sinh x$. Putting $\mathrm{i}x$ into $\cosh z = \frac{1}{2}(\mathrm{e}^z + \mathrm{e}^{-z})$ and $\sinh z = \frac{1}{2}(\mathrm{e}^z - \mathrm{e}^{-z})$ gives $\cosh(\mathrm{i}x) = \cos x$ and $\sinh(\mathrm{i}x) = \mathrm{i}\sin x$.

Example. (a) Write $\sin 5\theta$ in powers of $s = \sin\theta$. (b) Write $\cos^3\theta$ as a sum of cosines of multiples of θ. (c) Evaluate $\sin(\mathrm{i}\ln 3)$.

(a) With $c = \cos\theta$, the imaginary terms of $(c + \mathrm{i}s)^5$ are $5c^4(\mathrm{i}s) + 10c^2(\mathrm{i}s)^3 + (\mathrm{i}s)^5 = \mathrm{i}(5c^4s - 10c^2s^3 + s^5)$. Comparing imaginary parts with $\cos 5\theta + \mathrm{i}\sin 5\theta$, and using $c^2 = 1 - s^2$,

$$\begin{aligned}\sin 5\theta &= 5(1-s^2)^2 s - 10(1-s^2)s^3 + s^5 \\ &= \boxed{16s^5 - 20s^3 + 5s}.\end{aligned}$$

(b) Put $u = \mathrm{e}^{\mathrm{i}\theta}$. Then

$$\cos^3\theta = \tfrac{1}{8}\left(u + u^{-1}\right)^3 = \tfrac{1}{8}\left(u^3 + u^{-3} + 3(u + u^{-1})\right) = \tfrac{1}{8}\left(2\cos 3\theta + 6\cos\theta\right) = \boxed{\tfrac{1}{4}\cos 3\theta + \tfrac{3}{4}\cos\theta}.$$

(c) $\sin(\mathrm{i}\ln 3) = \mathrm{i}\sinh(\ln 3) = \mathrm{i}\cdot\frac{1}{2}\left(3 - \frac{1}{3}\right) = \boxed{\frac{4}{3}\mathrm{i}}$.

Check. Test at an angle where nothing vanishes. At $\theta = \frac{\pi}{6}$, where $s = \frac{1}{2}$: $\sin\frac{5\pi}{6} = \frac{1}{2} = \frac{16}{32} - \frac{20}{8} + \frac{5}{2}$. At $\theta = \frac{\pi}{3}$: $\cos^3\frac{\pi}{3} = \frac{1}{8} = \frac{1}{4}(-1) + \frac{3}{4}\cdot\frac{1}{2}$. For (c), from the inverse Euler formula directly: $\sin(\mathrm{i}\ln 3) = \frac{1}{2\mathrm{i}}(\mathrm{e}^{-\ln 3} - \mathrm{e}^{\ln 3}) = \frac{1}{2\mathrm{i}}(\frac{1}{3} - 3) = \frac{4}{3}\mathrm{i}$.

Pitfalls

- Getting $(2\mathrm{i})^n$ wrong: $(2\mathrm{i})^2 = -4$ and $(2\mathrm{i})^3 = -8\mathrm{i}$.
- Comparing parts before the expansion is complete, or putting a term with i^2 in the imaginary part.
- Writing $\cos(\mathrm{i}x) = \mathrm{i}\cosh x$. The cosine of an imaginary number is real: $\cos(\mathrm{i}x) = \cosh x$.

In the notes. Unit 3: Multiple-angle identities, Unit 3: Sine and cosine as exponentials, and Unit 3: A link to the hyperbolic functions.

Practise it. the Review set on this method

3.5 Adding sinusoids with phasors

Problem. Write a sum of sinusoids of one angular frequency ω (not the root of unity of Method 3.3) as a single sinusoid $R\cos(\omega t + \alpha)$.

Method

1. Write every term as $A\cos(\omega t+\varphi)$ with $A>0$. A sine becomes a cosine by $\sin x=\cos(x-\frac{\pi}{2})$, and a minus sign by $-\cos x=\cos(x+\pi)$.
2. Each term's phasor is $P=Ae^{i\varphi}$.
3. Add the phasors in Cartesian form. If they add up to 0, the signal is identically 0 and has no phase.
4. Otherwise find the modulus R and the argument α of the sum, quadrant first.
5. The sum is $R\cos(\omega t+\alpha)$. If the question asks for $R\cos(\omega t-\beta)$, then $\beta=-\alpha$.

Example. Write $4\cos(\omega t+\frac{5\pi}{6})-4\sin\omega t$ as a single sinusoid.

By both rules of step 1, $-\sin x=-\cos(x-\frac{\pi}{2})=\cos(x+\frac{\pi}{2})$, so the phasors are $P_1=4e^{i5\pi/6}=-2\sqrt{3}+2i$ and $P_2=4e^{i\pi/2}=4i$. Their sum is $-2\sqrt{3}+6i$, in the second quadrant, with reference angle $\arctan\frac{6}{2\sqrt{3}}=\arctan\sqrt{3}=\frac{\pi}{3}$. So

$$R=\sqrt{12+36}=4\sqrt{3},\qquad \alpha=\pi-\tfrac{\pi}{3}=\tfrac{2\pi}{3},$$

and the sum is $\boxed{4\sqrt{3}\cos(\omega t+\frac{2\pi}{3})}$, with amplitude about 6.93.

Check. At $\omega t=0$: $4\cos\frac{5\pi}{6}=-2\sqrt{3}=4\sqrt{3}\cos\frac{2\pi}{3}$. At $\omega t=\frac{\pi}{2}$, where the sine term counts in full: $4\cos\frac{4\pi}{3}-4=-6=4\sqrt{3}\cos\frac{7\pi}{6}$.

Pitfalls

- Adding the amplitudes: $4+4=8$ is too big unless the phases agree.
- Turning a sine the wrong way: $\sin x=\cos(x-\frac{\pi}{2})$, not $\cos(x+\frac{\pi}{2})$.
- Reading α off arctan when the phasor sum is in the second or third quadrant: here $\arctan\frac{6}{-2\sqrt{3}}=-\frac{\pi}{3}$ points the wrong way.

In the notes. Unit 3: Application: phasors and AC analysis.

Practise it. the Review set on this method